BY ORDER OF THE COMMANDER
AIR FORCE TEST CENTER

AIR FORCE TEST CENTER
INSTRUCTION 99-103

31 MAY 2018
Certified Current, 16 March 2022
Test and Evaluation

AFTC DISTRIBUTED
TEST OPERATIONS

COMPLIANCE WITH THIS PUBLICATIONS IS MANDATORY

ACCESSIBILITY: Publications and forms are available on the e-Publishing website at **www.e-Publishing.af.mil** for downloading or ordering

RELEASABILITY: There are no releasability restrictions on this publication

---

OPR: AFTC/ENT

Certified by: AFTC/EN
(Mr. Mallory Knight)
Pages: 11

This Air Force Test Center Instruction (AFTCI) implements Air Force Instruction 99-103, *Capabilities-Based Test and Evaluation*. It provides the requirements for Distributed Test Operations (DTO) with the AFTC Enterprise. This instruction applies to all AFTC Wing and Complex organizations. This publication may be supplemented at any level, but all direct supplements must be routed to the Office of Primary Responsibility (OPR) of this publication for coordination prior to certification and approval. Waiver authority for this instruction is the Air Force Test Center Commander. Refer recommended changes and questions about this publication to the OPR listed above using the AF Form 847, *Recommendation for Change of Publication*; route AF Forms 847 from the field through the appropriate chain of command. Ensure that all records created as a result of processes described in this publication are maintained in accordance with Air Force Manual (AFMAN) 33-363, *Management of Records*, and disposed of in accordance with Air Force Records Information Management System (AFRIMS) Records Disposition Schedule (RDS). The use of the name or mark of any specific manufacturer, commercial product, commodity, or service in this publication does not imply endorsement by the Air Force.

Chapter 1— PROGRAM OVERVIEW **3**

1.1. Overview. 3

Chapter 2— ROLES AND RESPONSIBILITIES SECTION **4**

2.1. Roles and Responsibilities Section. 4

2.2. LDTO Responsibilities: 4

2.3. ETO Responsibilities: 4

2.4. PTO Responsibilities 5

Chapter 3— DTO ACTIVITIES **6**

3.1. DTO Assessment Process. 6

3.2. Wing/Complex DTO Assessment Reviews. 6

3.3. AFTC DTO Assessment Review. 6

3.4. AFTC DTO Handbook. 6

Attachment 1— GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION **7**

Attachment 2— DTO ASSESSMENT TEMPLATE **9**

Chapter 1

PROGRAM OVERVIEW

1.1. Overview. DTO is defined as any and all operations where an AFTC Wing or Complex can leverage capabilities or resources existing outside of that organization. The intent of DTO is to facilitate faster and more efficient test that is uninhibited or not constrained by Wing-specific resources or processes. All AFTC test organizations are directed to employ DTO processes and methodologies whenever a planned test requires capabilities or resources beyond, or in excess of, those existing within their Wing/Complex (taking into account the customers cost and schedule constraints). When a required capability exists outside their Wing/Complex, organizations should not duplicate that capability (unless increased capacity is required for the AFTC), rather utilize DTO as the preferred approach. In cases where the Wing/Complex possesses all required capabilities and resources, but where scheduling conflicts may result in delays in test execution, DTO will be explored as a way to minimize delays.

Chapter 2

ROLES AND RESPONSIBILITIES SECTION

2.1. Roles and Responsibilities Section. The Air Force Test Center (AFTC) is one of Air Force Materiel Command's (AFMC's) Lead Developmental Test and Evaluation Organizations (LDTOs). The AFTC Enterprise conducts AFMC's Developmental Test and Evaluation (DT&E) mission through its three Executing Test Organizations (ETOs): the 412th Test Wing, the 96th Test Wing, and the Arnold Engineering Development Complex (AEDC). The ETOs are responsible for the planning, provision, execution, data analysis and reporting for test programs, projects or efforts assigned to them to conduct. The ETO may require additional resources beyond its local capability or capacity, and will partner with Participating Test Organizations (PTOs) that may be another AFTC wing or complex or an organization outside the AFTC Enterprise. This ETO/PTO partnership is the essence of DTO.

2.2. LDTO Responsibilities: The AFTC oversees its confederation of DT&E organizations (96 TW, 412 TW, and AEDC). As the OPR, AFTC/ENT serves as the AFTC DTO functional and will:

2.2.1. Develop and maintain AFTC Enterprise guidance for LDTO, ETO, PTO roles and responsibilities and the conduct of DTO.

2.2.2. Develop and maintain a comprehensive AFTC enterprise resource list identifying shareable test capabilities and provide a resource for test teams to find and utilize them.

2.2.3. Analyze DTO related information to enable appropriate investment in capabilities, resources, and processes to further support DTO operations.

2.2.4. Facilitate the AFTC Engineering Council's review of the DTO Assessment data on a semi-annual basis and brief the AFTC Council as necessary/directed.

2.2.5. Capture lessons learned and DTO collaboration roadblocks to sharing resources across the center and identify potential enhancements to the DTO process.

2.2.6. Update and maintain the AFTC DTO Handbook.

2.3. ETO Responsibilities: Wing or Complex will:

2.3.1. Ensure local processes are in place to facilitate and support DTO activities.

2.3.2. Plan and execute test missions making appropriate use of AFTC Enterprise resources to meet the program's schedule, objective, and cost goals.

2.3.3. Complete a DTO Assessment for every Statement of Capability (SOC) or SOC re-baseline and provide a copy to the AFTC DTO functional.

2.3.4. Establish Wing/Complex DTO Assessment review process.

2.3.5. Technical Directors will participate in AFTC Engineering Council reviews of the DTO Assessment data on a semi-annual basis.

2.3.6. Assist in the development and maintenance of the enterprise resource list to identify shareable test capabilities and provide a resource to test teams.

2.3.7. Assign an office of primary responsibility (OPR) for updating the enterprise resource list.

2.4. PTO Responsibilities : As PTO, AFTC test organizations providing DTO support will:

2.4.1. Provide ETO requested information required to complete the DTO Assessment.

2.4.2. Provide a Transfer of Workload (TOW) Letter of Agreement (LOA) to the ETO in support of the SOC generation process IAW AFTCI 99-100, *Statements of Capability*.

Chapter 3

DTO ACTIVITIES

3.1. DTO Assessment Process. The DTO Assessment is the first step in the process, whereby test programs assess the use and/or applicability of DTO when developing test strategies for Statement of Capability (SOC) and SOC re-baselining activities, utilizing the DTO Assessment template and samples provided in **Attachment 2**. The process is complete when the DTO Assessment has been sent to the AF Test Center LDTO Workflow for upload on the AFTC LDTO/ETO SharePoint site.

3.2. Wing/Complex DTO Assessment Reviews. The local DTO Assessment review is accomplished by the wing or complex, providing local validation of the program's decision regarding DTO utilization. The reviews informs leadership of trends in DTO utilizations and highlights capability limitations at the local level.

3.3. AFTC DTO Assessment Review. The AFTC DTO assessment review analyzes DTO utilization and limitation information to enable appropriate investment in capability, resources, and processes to support and enable DTO operations across the AFTC enterprise. These assessments will provide an understanding of how current capabilities are being used for DTO and why decisions are being made not to use DTO. They will also identify limitations that are currently preventing the use of DTO. AFTC/ENT, as the AFTC DTO functional, will generate metric data for review at the AFTC Engineering Council on a semi-annual basis and brief the AFTC Council as necessary or directed.

3.4. AFTC DTO Handbook. The AFTC DTO Handbook is intended to be a guide to conducting DTO testing and includes details about; DTO Assessment process, how to handle changes to charging practices, disposition of assets or facilities, and determination of program risk and uncertainty. Updates to the AFTC DTO Handbook will be accomplished by the AFTC DTO functional as necessary.

DAVID A. HARRIS, Major General, USAF
Commander

Attachment 1

GLOSSARY OF REFERENCES AND SUPPORTING INFORMATION

References

AFI 99-103, *Capabilities-Based Test and Evaluation,* 6 April 2017

AFMAN 33-363, *Management of Records,* 1 March 2008

AFTCI 99-100, *Statements of Capability,* 15 October 2015

Adopted Forms

AF Form 847, *Recommendation for Change of Publication*

Abbreviations and Acronyms

AEDC—Arnold Engineering Development Complex

AF—Air Force

AFB—Air Force Base

AFMC—Air Force Materiel Command

AFTC—Air Force Test Center

AFTCI—Air Force Test Center Instruction

DT&E—Developmental Test and Evaluation

DTO—Distributed Test Operations

ETO—Executing Test Organization

IAW—In Accordance With

LDTO—Lead Developmental Test and Evaluation Organization

LOA—Letter of Agreement

OPR—Office of Primary Responsibility

PTO—Participating Test Organization

SOC—Statement of Capability

TOW—Transfer of Workload

TW—Test Wing

Terms

Executing Test Organization (ETO) —This is the primary test site/organization accomplishing DT for the AFTC when the AFTC has been identified as the LDTO.

Lead Developmental Test and Evaluation Organization (LDTO) —The LDTO is nominated by the ITT and approved by the PEO. The LDTO functions as the lead integrator for a program's DT&E activities. It is separate from the program office, but supports the PM and ITT in a

provider-customer relationship with regard to scope, type and conduct of required DT&E. The LDTO assists the CDT with oversight of contractor DT&E results and managing studies, analyses and program documentation from the requirements, acquisition and cyber test communities. The LDTO is selected from the list of qualified candidates published by AFMC and AFSPC. The AFTC is an LDTO, as identified in AFMC Form 42 instructions.

Participating Test Organization (PTO)—Any test organization required to act in a supporting role to the ETO or LDTO by providing specific T&E data or resources for a T&E program or activity.

Re-baseline —An update to cost and/or schedule agreed to by the PM, Wing/Complex leadership and the customer. Although sites may use other terms, if the cost and/or schedule commitments have been changed, then the AFTC considers it a re-baseline and it must be reported as such.

Wing/Complex —Generic reference to any of the AFTC components: 412th Test Wing, 96th Test Wing, or AEDC**.**

Attachment 2

DTO ASSESSMENT TEMPLATE

Figure A2.1. DTO Assessment Template.

DTO Assessment
Every test effort is required to accomplish a DTO Assessment to consider if Distributed Test Operations can facilitate faster, more efficient and cost effective test that is uninhibited or constrained by Wing-Specific resources or processes.
Test Effort Title: **Assessment Date**:
Assessor (Name/Org/Ph#):
After considerations of the test effort's resource requirements and the AFTC's test capabilities, does the test effort plan to utilize DTO? (Include external to AFTC resources, as applicable.)
1. Yes, DTO will be utilized to support this test effort. (Completely answer questions a and b)

a. Please identify which category of DTO capability/resource to be utilized during this test effort. (Select all that apply to your test effort from the list below)
i. Manpower
ii. Facility
iii. Range
iv. Other, briefly identify:

b. Are there any limitations to existing DTO capability, resource, or process to be utilized? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.
i. Manpower
ii. Facility
iii. Range
iv. Was cost a determining factor, briefly explain:
v. Other, briefly identify:

2. No, DTO will not be utilized to support this test effort. (Completely answer questions a, b and c)

a. Does the ETO possess all the necessary capability/resources to accomplish this test effort? (Briefly explain your rationale.)

b. Is the test effort planning to enhance or expand any existing capability or capacity at your Wing to support this effort? (If so, briefly describe the enhancement/expansion.)

c. Are there any resources the test effort required but did not exist or has significant limitations that prevented utilization, internal to the Wing or at the AFTC Enterprise? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.
i. Manpower
ii. Facility
iii. Range
iv. Was cost a determining factor, briefly explain:
v. Other, briefly identify

DTO Assessment (Sample 1 – DTO utilized)

DTO Assessment
Every test effort is required to accomplish a DTO Assessment to consider if Distributed Test Operations can facilitate faster, more efficient and cost effective test that is uninhibited or constrained by Wing-Specific resources or processes.
Test Effort Title: *F-16 Weapons* **Test Assessment Date**: *DD/MM/YYYY*
Assessor (Name/Org/Ph#): *Joe Tester / 96TW / DSN# XXX-XXXX*
After considerations of the test effort's resource requirements and the AFTC's test capabilities, does the test effort plan to utilize DTO? (Include external to AFTC resources, as applicable.)

1. Yes, DTO will be utilized to support this test effort. (Completely answer questions a and b)

a. Please identify which category of DTO capability/resource to be utilized during this test effort. (Select all that apply to your test effort from the list below)
i. Manpower: *412TW/TENG, 412TW Chase Aircraft Aircrew*
ii. Facility: *Ramp space for deployed aircraft and support team*
iii. Range: *China Lake*
iv. Other, briefly identify:
- *Business Tanker Support near China Lake*
- *Safety Chase Support from Edwards AFB*

b. Are there any limitations to existing DTO capability, resource, or process to be utilized? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.
i. Manpower
ii. Facility
iii. Range: *Due to accelerated program requirements, data connection requirements could not be met with existing capabilities between Edwards AFB and Eglin AFB. Program had to deploy all mission essential personnel.*
iv. Was cost a determining factor, briefly explain:
v. Other, briefly identify:

2. No, DTO will not be utilized to support this test effort. (Completely answer questions a, b and c)
a. Does the ETO possess all the necessary capability/resources to accomplish this test effort? (Briefly explain your rationale.)
b. Is the test effort planning to enhance or expand any existing capability or capacity at your Wing to support this effort? (If so, briefly describe the enhancement/expansion.)
c. Are there any resources the test effort required but did not exist or has significant limitations that prevented utilization, internal to the Wing or at the AFTC Enterprise? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.
i. Manpower
ii. Facility
iii. Range
iv. Was cost a determining factor, briefly explain:
v. Other, briefly identify:

DTO Assessment (SAMPLE 2 – DTO not utilized)

DTO Assessment

Every test effort is required to accomplish a DTO Assessment to consider if Distributed Test Operations can facilitate faster, more efficient and cost effective test that is uninhibited or constrained by Wing-Specific resources or processes.

Test Effort Title: *Engine-X Propulsion* **Test Assessment Date**: *DD/MM/YYYY*

Assessor (Name/Org/Ph#): Jane Tester / AEDC / DSN# XXX-XXXX

After considerations of the test effort's resource requirements and the AFTC's test capabilities, does the test effort plan to utilize DTO? (Include external to AFTC resources, as applicable.)

1. Yes, DTO will be utilized to support this test effort. (Completely answer questions a and b)

a. Please identify which category of DTO capability/resource to be utilized during this test effort. (Select all that apply to your test effort from the list below)

i. Manpower
ii. Facility
iii. Range
iv. Other, briefly identify:

b. Are there any limitations to existing DTO capability, resource, or process to be utilized? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.

i. Manpower
ii. Facility
iii. Range
iv. Was cost a determining factor, briefly explain:
v. Other, briefly identify:

2) No, DTO will not be utilized to support this test effort. (Completely answer questions a, b and c)

a. Does the ETO possess all the necessary capability/resources to accomplish this test effort? (Briefly explain your rationale.)

- *All resources required for testing are possessed by AEDC and AEDC has the only Facility capable of supporting this propulsion test effort.*

b. Is the test effort planning to enhance or expand any existing capability or capacity at your Wing to support this effort? (If so, briefly describe the enhancement/expansion.)

- *Test will require a modification to Facility X and it is the only facility in the AFTC Enterprise with the capability to support this effort.*

c. Are there any resources the test effort required but did not exist or has significant limitations that prevented utilization, internal to the Wing or at the AFTC Enterprise? (Briefly explain.) This data will support AFTC's evaluation of future capability enhancements.

i. Manpower
ii. Facility
iii. Range
iv. Was cost a determining factor, briefly explain
v. Other, briefly identify: *No other limitations have been identified.*